Muminul Islam

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Education

BRAC University 2026 – Present
M.Sc. in Computer Science and Engineering *Dhaka, Bangladesh*

United International University 2020 – 2025
B.Sc. in Computer Science and Engineering *Dhaka, Bangladesh*

- **Undergraduate Thesis:** Developed a 5,000+ sentence Bangla–Chittagonian parallel dataset and fine-tuned Transformer-based models for low-resource machine translation.

Research Experience

Academic Research Collaborator November 2025 – September 2026
Supervised by Dr. Md Adnan Arefeen, North South University

- Co-authored *"RE-SciVQA: Improving Scientific VQA with Context-Relevant Retrieval and Lightweight MLLM Adaptation,"* currently under review at **IEEE Access**. [Manuscript]
- Developed a context-aware retrieval-augmented generation pipeline with query expansion and parameter-efficient QLoRA adaptation for scientific figure question answering.
- Contributed to evaluation of multimodal models including **ChartGemma** and **UniChart**, with emphasis on complex scientific figures and closed-ended visual questions.

Professional Experience

Explainable LLM and Interpretable Technology Ensemble Research Lab September 2026 – Present
Researcher (Part-Time) *Remote – Queens, NY, USA*

Harris Digital (Harris Web Works) June 2025 – November 2025
Trainee Software Engineer *Dhaka, Bangladesh*

- Introduced AI-assisted development and prompt-engineering workflows using tools such as ChatGPT, OpenAI Codex, and GitHub Copilot to accelerate prototyping and development.

Competitive Programming & Achievements

- **Champion** — UIU Intra-University AI Contest, Fall 2022; developed a multiclass classification model to predict medical conditions from research-paper abstracts.
- **ICPC Asia Dhaka Regional 2024**: Rank 135/309 onsite and Rank 172/2400+ in the online preliminary; Honourable Mention in 2021, 2022, and 2023.
- **Inter-University Programming Contests**: Rank 7 (UAP Collaborative Contest 2024), Rank 8 (UIU Coders Combat 3.0), Rank 50 (KUET IUPC 2025), Rank 52 (SUST IUPC 2024), and Rank 53 (IUT IUPC 2024).
- Solved **800+ algorithmic problems** across **150+ contests** on Codeforces (B4TMAN, max rating 1332), VJudge, and other online judges.
- Awarded the **UIU Merit Scholarship twice** for academic performance and placed **3rd Runner-up** at the System Analysis and Design Lab Project Show.

Technical Skills

Programming: C, C++, Python, C#, TypeScript

AI/ML: PyTorch/ML workflows, NumPy, Pandas, scikit-learn, TensorFlow, Transformers, RAG, QLoRA

Software Engineering: ASP.NET Core, React, PHP, REST APIs, MVC architecture

Databases & Tools: PostgreSQL, MySQL, Git, GitHub, Postman, Linux CLI, Bash

Selected Projects

- **E-Commerce Inventory API** | *ASP.NET Core, EF Core, JWT, PostgreSQL*
  Developed a secure .NET 8 REST API with JWT/BCrypt authentication, product/category CRUD, pagination, search, soft deletes, repository-pattern data access, and PostgreSQL support. [GitHub]
- **Codewar** | *PHP, MySQL, Bootstrap*
  Developed a competitive-programming platform with practice problems and analytics dashboards. [GitHub]
- **LinguaLearner** | *React Native, Express.js, TypeScript, MongoDB*
  Built a cross-platform language-learning application with quizzes, blogs, and secure user authentication. [GitHub]
- **Cinephiles** | *PHP, MySQL, JavaScript*
  Developed a movie platform featuring personalised recommendations, watch-party rooms, and community interaction. [GitHub]

Research Interests

- Generative and Multimodal AI, Large Language Models, Retrieval-Augmented Generation, AI for Software Engineering and Code Intelligence, Agentic and Trustworthy AI, and Efficient Model Adaptation.